

WEEK-3

Java programming -Java Buzzwords, Data types, Variables, Operators, Control structures, Arrays, Console input and output

Sample Program:

1. Sample Java program to check whether the given number is prime palindrome or not. CO1.
2. Write a java program which reads marks of five subjects through command line and print the total and average.

Algorithm for the code:

Algorithm: Prime Palindrome

Step 1: Start.

Step 2: Read an integer n from the user.

Step 3: Check whether n is prime using isPrime(n).

To check prime:

1. If $n \leq 1$, return false.
2. Start a loop from $i = 2$ to $n / 2$.
3. Check $n \% i$.
4. If $n \% i == 0$, the number is not prime, so return false.
5. If no divisor is found, return true.

Step 4: Check whether n is a palindrome using isPalindrome(n).

To check palindrome:

1. Store the original number in original.
2. Set reversed = 0.
3. Extract the last digit using $\text{num} \% 10$.
4. Add the digit to reversed.
5. Remove the last digit using $\text{num} / 10$.
6. Repeat until num becomes 0.
7. Compare original and reversed.
8. If they are equal, return true; otherwise return false.

Step 5: Check both results using:

isPrime(n) && isPalindrome(n)

Step 6: If both are true, display:

n is a Prime Palindrome.

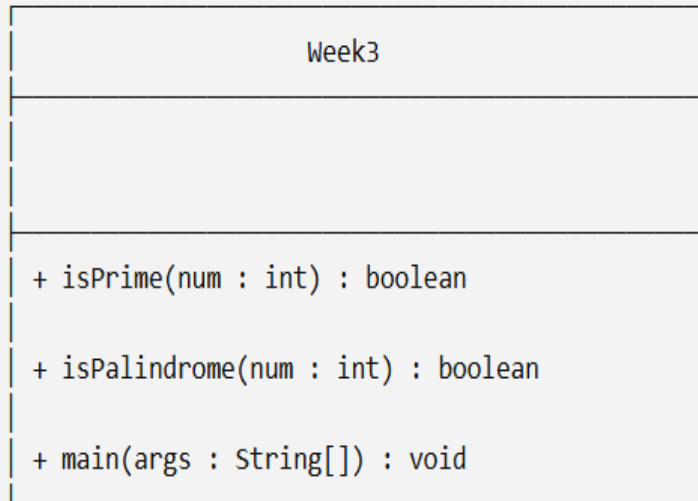
Otherwise, display:

n is NOT a Prime Palindrome.

Step 7: Close the Scanner.

Step 8: Stop.

3.UML diagram for the code:



4.Code:

```
import java.util.Scanner;
```

```
public class Week3 {
```

```
    public static boolean isPrime(int num) {
```

```
        if (num <= 1) return false;
```

```
        for (int i = 2; i <= num/2; i++) {
```

```
            if (num % i == 0) return false;
```

```
        }
```

```
        return true;
```

```
    }
```

```
    public static boolean isPalindrome(int num) {
```

```
        int original = num, reversed = 0;
```

```
        while (num > 0) {
```

```
            int digit = num % 10;
```

```

        reversed = reversed * 10 + digit;

        num /= 10;
    }

    return original == reversed;
}

```

```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a number: ");

    int n = sc.nextInt();

    if (isPrime(n) && isPalindrome(n)) {

        System.out.println(n + " is a Prime Palindrome.");

    } else {

        System.out.println(n + " is NOT a Prime Palindrome.");

    }

    sc.close();

}
}

```

5.output:

```

Enter a number: 121
121 is NOT a Prime Palindrome.
PS C:\Users\Harsha Vardhan\Desktop\java> ^C
PS C:\Users\Harsha Vardhan\Desktop\java>
PS C:\Users\Harsha Vardhan\Desktop\java> cd 'c:\Users\Harsha Vardhan\Desktop\java'; & 'c:\Program Files\Java\jdk-25.0.4\bin\java.exe' '-X
X:+ShowCodeDetailsInExceptionMessages' '-cp' 'c:\Users\Harsha Vardhan\AppData\Roaming\Code\User\workspaceStorage\f8641d91bba67b5feb3c9719e0895
bbc\redhat.java\jdt_ws\java_61d938da\bin' 'Week3'
Enter a number: 131
131 is a Prime Palindrome.
PS C:\Users\Harsha Vardhan\Desktop\java>

```